Maitri – A Digital Companion for Women's Health and Well-Being

A PROJECT REPORT

Submitted by
Aashna Goyal (24BCS10536)
Ravleen Kaur (24BCS10498)
Karan Juneja(24BCS10791)
Pratik(24BCS10403)

in partial fulfillment for the award of the degree of

Bachelor of Engineering

IN

Computer Science and Engineering

Chandigarh University

January 2026

BONAFIDE CERTIFICATE

Certified that this project report **"**Maitri – A Digital Companion for Women's Health and Well-Being**"** is the bonafide work of "*Aashna Goyal (24BCS10536), Ravleen Kaur (24BCS10498), Karan Juneja(24BCS10791) and Pratik(24BCS10403)***"** who carried out the project work under my/our supervision.

SIGNATURE
Dr. Jaspreet Singh Batth

SIGNATURE
Neeraj Sharma

HEAD OF THE DEPARTMENT

CSE - 2nd year

University Institute of Engineering

Submitted for the project viva-voce examination held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

TABLE OF CONTENTS

List of Figures....7

List of Tables.... 8

List of Standards....9

CHAPTER 1. INTRODUCTION....11

1.1. Identification of Client/ Need/ Relevant Contemporary issue.... 11

1.2. Identification of Problem.... 11

1.3. Identification of Tasks....11

1.4. Timeline.... 11

1.5. Organisation of the Report.... 11

CHAPTER 2. LITERATURE REVIEW/BACKGROUND STUDY....12

2.1. Timeline of the reported problem.... 12

2.2. Existing solutions....12

2.3. Bibliometric analysis.... 12

2.4. Review Summary....12

2.5. Problem Definition....12

2.6. Goals/Objectives.... 12

CHAPTER 3. DESIGN FLOW/PROCESS.... 13

3.1. Evaluation & Selection of Specifications/Features....13

3.2. Design Constraints.... 13

3.3. Analysis of Features and finalisation subject to constraints.... 13

3.4. Design Flow.... 13

3.5. Design selection.... 13

3.6. Implementation plan/methodology.... 13

CHAPTER 4. RESULTS ANALYSIS AND VALIDATION **14**

4.1. Implementation of solution 14

CHAPTER 5. CONCLUSION AND FUTURE WORK **15**

5.1. Conclusion 15

5.2. Future work 15

REFERENCES **16**

APPENDIX **17**

1. Plagiarism Report 17

2. Design Checklist 17

USER MANUAL **18**

List of Standards (Mandatory For Engineering Programs)

Standard	Publishing Agency	About the standard	Page no
IEEE 802.11	IEEE	IEEE 802.11 is part of the IEEE 802 set of local area network (LAN) technical standards and specifies the set of media access control (MAC) and physical layer (PHY) protocols for implementing wireless local area network (WLAN) computer communication.	Mention page nowhere standard is used

Note: Text in Red is presented as an example (replace with relevant information)

INTRODUCTION

1.1. Identification of Client /Need / Relevant Contemporary issue

- **Client**: Women and adolescent girls who use digital platforms for menstrual tracking, fertility awareness, mental wellness, and overall health management.
- **Need Identification:** Although applications like Fertility Friend, Flo Health, Clue, and Ovia exist, many users face issues such as inaccurate cycle prediction, poor user experience, misleading alerts, excessive ads, high pricing, and serious data privacy concerns. These problems indicate a clear need for a more reliable, private, and women-centric digital health solution.
- **Justification of the Problem:** Reports from global health organisations highlight gaps in women's reproductive and mental healthcare access. Studies show that average-based cycle prediction methods are often inaccurate, especially for irregular cycles. User reviews and surveys of existing apps report false pregnancy alerts, inability to process advanced fertility data (BBT, OPK), and anxiety-inducing content, proving that the problem is real and unresolved.
- **Survey-Based Evidence:** Feedback collected from users of existing women's health apps reveals dissatisfaction with data security, lack of personalisation, intrusive advertisements, and poor emotional support, justifying the need for an improved platform.
- **Contemporary Relevance:** Women's digital health privacy is a major contemporary issue, with reports from data protection and health agencies warning about the misuse of sensitive reproductive data. This highlights the urgent need for secure, trustworthy, and women-only health platforms.

1.2. Identification of Problem

Women face increasing difficulties in managing their reproductive, mental, and overall health through existing digital health platforms due to inaccurate health predictions, poor user experience, and a lack of personalisation. Many applications provide misleading or anxiety-inducing information, especially for users with irregular cycles, and fail to integrate physical and mental health support effectively. Concerns regarding data privacy, misuse of sensitive health information, excessive advertisements, and high subscription costs further reduce user trust. Additionally, the absence of safe, anonymous, and women-only digital spaces limits open discussion and reliable access to health guidance.

1.3. Identification of Tasks

1. **Problem Identification and Requirement Analysis**
This task involves studying existing women's health platforms, identifying user pain points through surveys and secondary research, and defining functional and non-functional requirements. It also includes understanding user expectations related to accuracy, privacy, usability, and mental wellness.
2. **System Design and Planning**
This phase focuses on designing the overall system architecture, defining the technology stack, planning data flow, and creating UI/UX wireframes. Security, accessibility, and scalability considerations are also addressed at this stage.
3. **Development and Implementation**
This task includes building the core modules such as user authentication, cycle and mood tracking, community support features, content management, and data protection mechanisms. Each feature is developed and integrated according to the defined requirements.
4. **Testing and Validation**
Testing is performed to verify functionality, accuracy, usability, and security. This includes unit testing, integration testing, and user acceptance testing to ensure the system meets user needs and performs reliably.
5. **Evaluation and Documentation**
The final task involves analysing results, evaluating system performance against objectives, identifying limitations, and documenting the complete development process, outcomes, and future scope

1.4. Timeline

Define the timeline (preferably using a Gantt chart)

1.5. Organisation of the Report

Give a brief of what should be expected in each of the chapters

LITERATURE REVIEW/BACKGROUND STUDY

2.1. Timeline of the reported problem

As investigated throughout the world, when the problem is identified, documentary proof of the incidents.

2.2. Existing solutions

Several digital platforms, such as Flo Health, Clue, Ovia Health, and Fertility Friend, have been developed to support women's reproductive and menstrual health. These applications provide features like cycle tracking, fertility prediction, symptom logging, and access to health-related information, which have improved awareness and accessibility for many users. By offering digital tools for self-monitoring, these platforms represent an important step toward integrating technology with women's healthcare.

However, closer analysis reveals significant limitations in existing solutions. Many applications rely on average-based cycle prediction methods, leading to inaccurate results, especially for women with irregular cycles. Some platforms, such as Fertility Friend, have interfaces that are not intuitive for first-time users. Additionally, users have reported false pregnancy alerts and misleading health tips, which can cause confusion and anxiety rather than reassurance.

Further concerns include limited support for advanced fertility inputs such as basal body temperature (BBT) and ovulation predictor kit (OPK) data, reducing predictive accuracy. Issues related to data privacy, potential misuse or scamming, excessive advertisements, and high subscription costs have also been widely reported. Moreover, the lack of women-only privacy controls and the presence of unnecessary or emotionally distressing content highlight gaps in emotional sensitivity and user trust within current platforms.

2.3. Bibliometric analysis

The literature also consistently reports several drawbacks, including poor usability for first-time users, misleading alerts, limited support for advanced fertility data such as BBT analysis. Existing research literature and documented studies on women's digital health applications show that most platforms emphasise menstrual cycle tracking and fertility prediction as their primary features. Commonly identified functionalities include period tracking, symptom monitoring, health notifications, and access to informational content related to reproductive health.

From an effectiveness perspective, these applications contribute positively to health awareness and self-monitoring among users. Studies indicate that digital tracking tools help users understand menstrual patterns and encourage proactive health management. However, the effectiveness of these platforms is limited due to reliance on average-based prediction models, which often result in inaccurate outcomes, particularly for women with irregular temperature (BBT) and ovulation predictor kit (OPK) inputs, and inadequate integration of mental health support. Additionally, concerns regarding data privacy, excessive advertisements, high subscription costs, and lack of secure, women-only access are frequently documented, highlighting the need for more reliable and user-centric solutions.

2.4. Review Summary

Existing literature and women's health applications mainly focus on menstrual cycle tracking and fertility prediction. While they support self-monitoring and awareness, generalised prediction models often lead to inaccuracies, especially for irregular cycles. Other issues include poor usability, misleading alerts, limited support for advanced fertility data (BBT, OPK), data privacy concerns, excessive ads, and a lack of emotionally sensitive content.

Significant gaps remain in integrating mental wellness, lifestyle guidance, and safe, women-only community spaces. Maitri addresses these gaps by providing accurate tracking, advanced data support, mental health recommendations, and a secure, holistic platform that combines physical, mental, and emotional health management.

2.5. Problem Definition

The problem at hand is the lack of a comprehensive, accurate, and secure digital platform for women's health that integrates physical, mental, and emotional well-being. Existing applications focus mainly on menstrual and fertility tracking but often provide inaccurate predictions, poor usability, misleading alerts, limited support for advanced fertility data (BBT, OPK), and insufficient mental wellness guidance.

What is to be done:

- Develop a user-friendly platform that offers accurate cycle and fertility tracking.
- Include mental health support, lifestyle recommendations, and community guidance.
- Ensure data privacy, women-only access, and an ad-free experience.

How it is to be done:

Implement advanced tracking algorithms using multi-parameter data.
Provide an interactive, secure interface with anonymous community support.
Integrate SOS and crisis support features along with reliable content.

What is not to be done:

- The platform will not provide medical diagnosis or treatment.
- It will not share user data with third parties.
- It will avoid unnecessary advertisements, misleading alerts, or depressive content.

2.6. Goals/Objectives

The main goal of this project is to develop Maitri, a secure, holistic digital platform for women's health that integrates physical, mental, and emotional well-being. The specific objectives are:

1. **Accurate Cycle and Fertility Tracking** – Implement multi-parameter algorithms to predict menstrual cycles and fertility with improved accuracy.
2. **Mental Health and Wellness Support** – Provide mood tracking, lifestyle recommendations, and access to reliable wellness content.
3. **Safe and Private Environment** – Ensure women-only access, anonymous community support, and strict data privacy.
4. **User-Friendly Interface** – Design an interactive, intuitive, and accessible interface for easy navigation and engagement.
5. **Ad-Free and Affordable Access** – Avoid unnecessary advertisements and provide cost-effective or free access to essential features.

   These objectives are concrete, measurable, and verifiable, forming clear milestones for the development and evaluation of the project.